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(54) **METHOD OF CONFIGURING AIR
ACCELERATOR**

(56) **References Cited**

U.S. PATENT DOCUMENTS

(71) Applicant: **Altria Client Services LLC**,
Richmond, VA (US)

1,845,969 A 2/1932 Hueber
1,853,863 A 4/1932 Hornbruch
(Continued)

(72) Inventor: **Dwight D. Williams**, Powhatan, VA
(US)

FOREIGN PATENT DOCUMENTS

(73) Assignee: **Altria Client Services LLC**,
Richmond, VA (US)

EP 0276628 A1 8/1988
EP 0982229 A1 3/2000
(Continued)

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OTHER PUBLICATIONS

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Primary Examiner — Jason K Niesz

(74) *Attorney, Agent, or Firm* — Harness, Dickey &
Pierce, P.L.C.

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(Continued)

(57) **ABSTRACT**

The method includes first configuring a first cross-sectional
flow area of a throat within a first air accelerator to discharge
a target quantity of a particulate material, the first air
accelerator including a body and an adjustable member, an
interior surface of the body defining a cavity containing the
adjustable member, the throat being at an air outlet of a
chamber defined between the body and the adjustable mem-
ber, the throat being near a discharge end of a singular bore
defined by the adjustable member, the first configuring
including axially moving the adjustable member along a first
longitudinal length of the cavity.

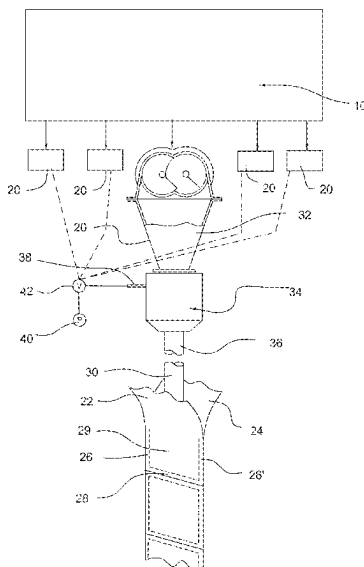
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CPC **B65B 37/14** (2013.01); **B65B 1/04**
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of application No. 15/973,860, filed on May 8, 2018, now Pat. No. 11,027,860, which is a continuation of application No. 13/546,649, filed on Jul. 11, 2012, now Pat. No. 9,963,253.

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(56) **References Cited**

U.S. PATENT DOCUMENTS

2,145,941	A	2/1939	Maxfield
2,385,897	A	10/1945	Waters
2,832,615	A	4/1958	Summers
2,862,529	A	12/1958	Rosenblatt et al.
2,915,866	A	12/1959	Bartlo
3,027,695	A	4/1962	Leasure
3,050,430	A *	8/1962	Gallagher A24D 3/0233 28/271
3,070,931	A	1/1963	Zwight
3,081,951	A	3/1963	Dyer et al.
3,173,188	A	3/1965	Wexler
3,380,393	A	4/1968	Nielsen
3,381,441	A	5/1968	Condo, Jr. et al.
3,402,446	A	9/1968	Benson
3,429,273	A	2/1969	Jones, Jr.
3,509,799	A	5/1970	Weis et al.
3,525,193	A	8/1970	Cutler et al.
3,637,447	A	1/1972	Berger
4,079,662	A	3/1978	Puccetti et al.
4,111,212	A	9/1978	Bakker
4,135,615	A	1/1979	Brackmann et al.
4,144,041	A	3/1979	Hou
4,160,526	A	7/1979	Flanagan
4,179,323	A	12/1979	Sigmon
4,228,900	A	10/1980	Lambach et al.
4,291,520	A	9/1981	Prince et al.
4,346,504	A	8/1982	Birk et al.
4,391,081	A	7/1983	Kovacs
4,468,845	A	9/1984	Harris
4,549,875	A	10/1985	Pryor
4,563,862	A	1/1986	McElvy
4,630,429	A	12/1986	Christine
4,703,765	A	11/1987	Paules et al.
4,757,668	A	7/1988	Klinkel et al.
4,807,814	A *	2/1989	Douche B05B 7/1486 417/174
4,860,959	A	8/1989	Handleman
4,869,048	A	9/1989	Boeckmann
5,012,629	A	5/1991	Rehman et al.
5,038,550	A	8/1991	Wirsig et al.

5,060,351	A	10/1991	Street
5,109,893	A	5/1992	Derby
5,174,088	A	12/1992	Focke et al.
5,247,779	A	9/1993	Wirsig et al.
5,279,098	A	1/1994	Fukuda
5,326,416	A	7/1994	Perrett
5,443,102	A	8/1995	Svensden
5,722,217	A	3/1998	Cloud
5,771,667	A	6/1998	McGregor et al.
5,873,680	A	2/1999	Huber et al.
5,937,614	A	8/1999	Watkins et al.
6,116,001	A *	9/2000	Kammler B65B 9/20 53/551

6,135,120	A	10/2000	Lofman et al.
6,170,768	B1	1/2001	Harmon et al.
6,220,000	B1 *	4/2001	Kammler B65B 37/14 53/511

6,237,308	B1	5/2001	Quintin et al.
6,261,030	B1	7/2001	Rassman et al.
6,301,859	B1	10/2001	Nakamura et al.
6,347,499	B1	2/2002	McGregor et al.
6,417,676	B1	7/2002	Schroder et al.
6,550,226	B1	4/2003	Gates et al.
6,631,605	B1	10/2003	Quintin et al.
6,732,897	B2	5/2004	DeHart et al.
6,749,027	B1	6/2004	Crabtree et al.
6,982,094	B2	1/2006	Sowden
7,003,931	B2	2/2006	Gates et al.
7,134,256	B2	11/2006	Kujubu et al.
7,708,504	B2	5/2010	Heckendorn et al.
11,027,860	B2	6/2021	Williams
2003/0222372	A1	12/2003	Ward
2004/0042857	A1	3/2004	DeHart et al.
2006/0081723	A1	4/2006	Turberville
2007/0012328	A1	1/2007	Winterson et al.
2007/0048097	A1	3/2007	Heckendorn et al.
2009/0272389	A1	11/2009	Atwell et al.
2010/0101189	A1	4/2010	Boldrini
2010/0163256	A1	7/2010	Williams
2011/0239591	A1	10/2011	Gruss et al.
2011/0277878	A1	11/2011	Williams
2011/0303232	A1	12/2011	Williams
2012/0000165	A1	1/2012	Williams

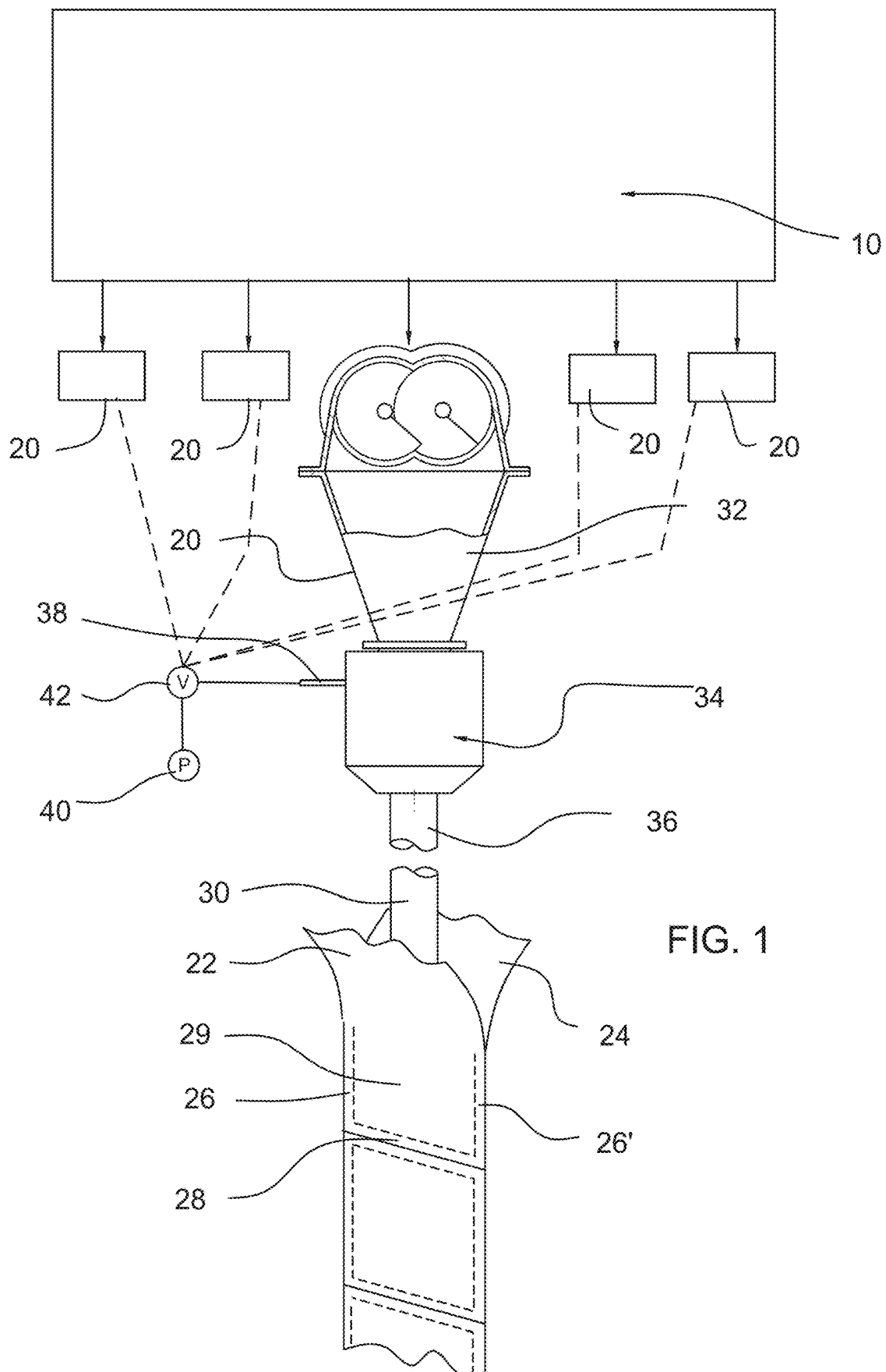
FOREIGN PATENT DOCUMENTS

EP	1637456	A1	3/2006
WO	WO-9008083	A1	7/1990
WO	WO-2009021855	A1	2/2009
WO	WO-2009047627	A1	4/2009
WO	WO-2010051826	A1	5/2010

OTHER PUBLICATIONS

International Search Report and Written Opinion mailed Sep. 28, 2012 for PCT/US2012/046237.

* cited by examiner



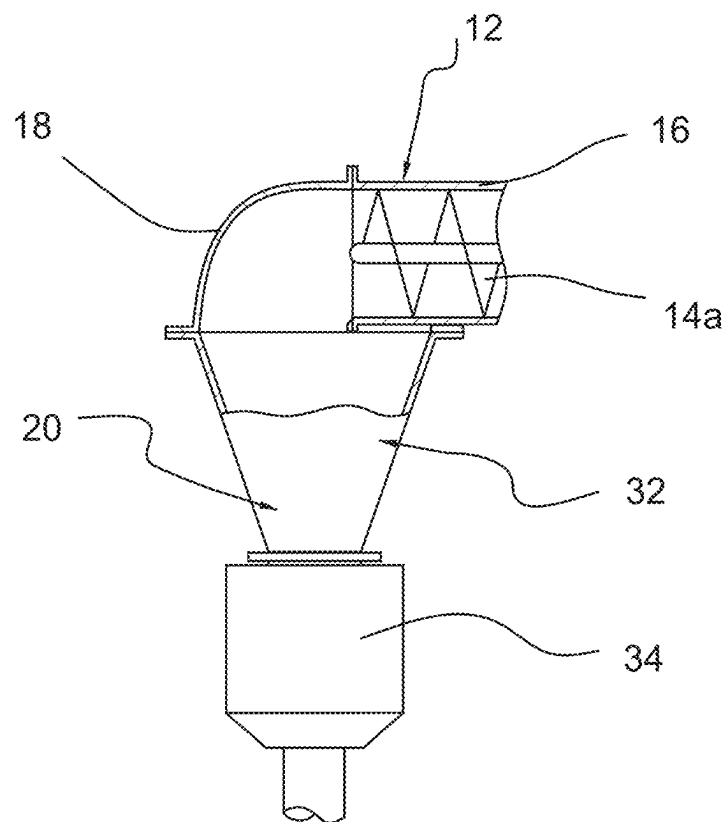
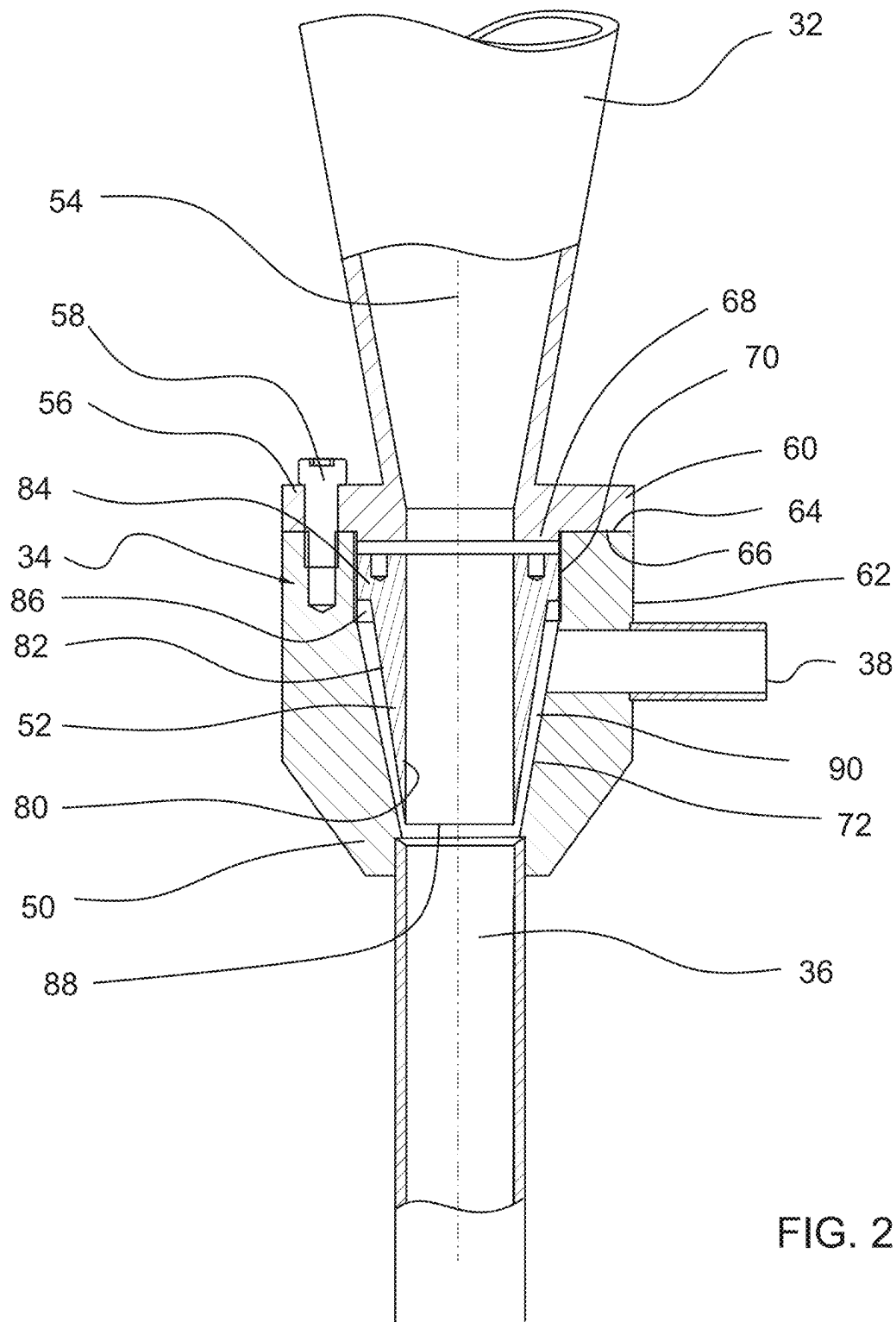


FIG. 1A



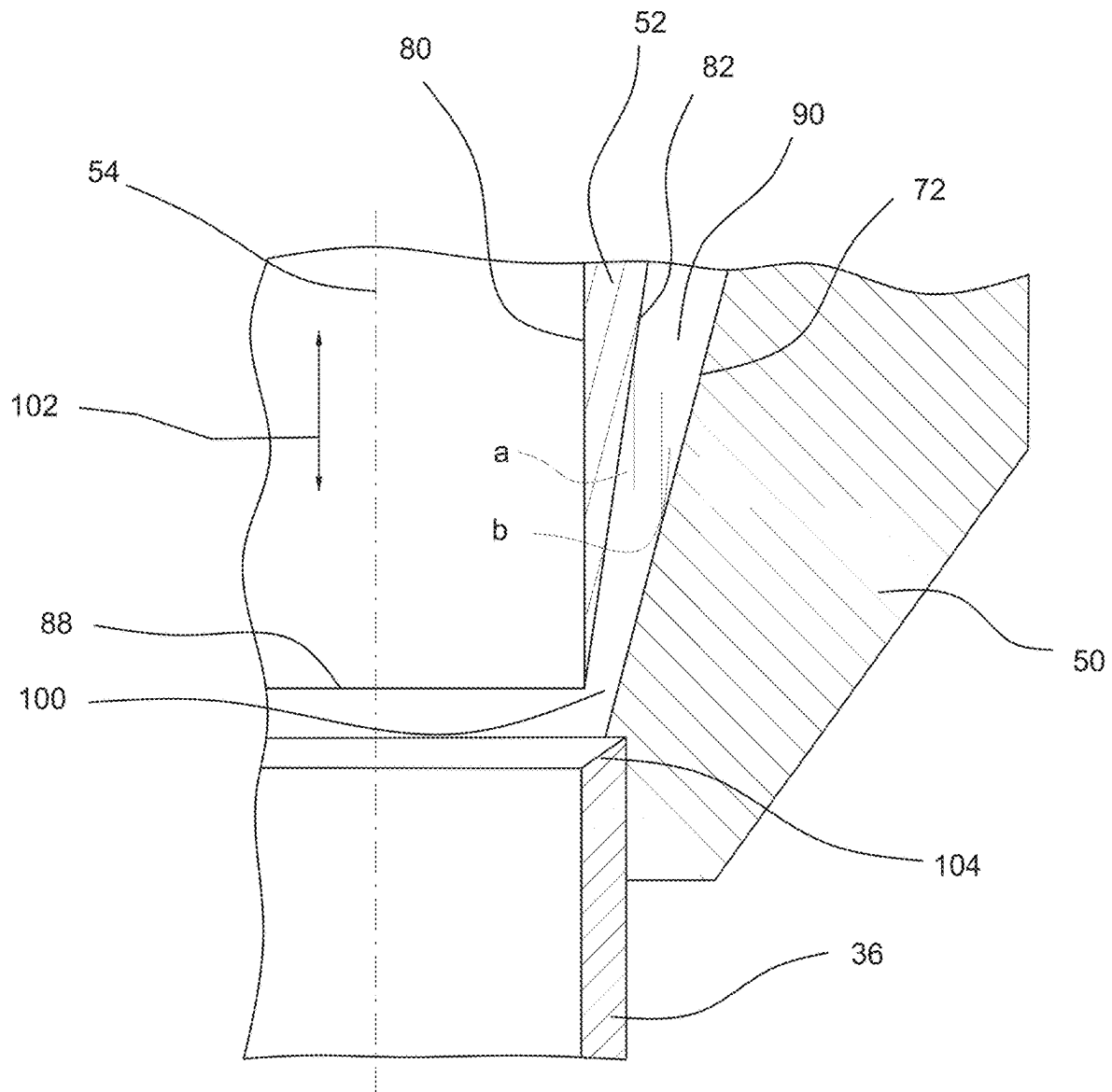


FIG. 3

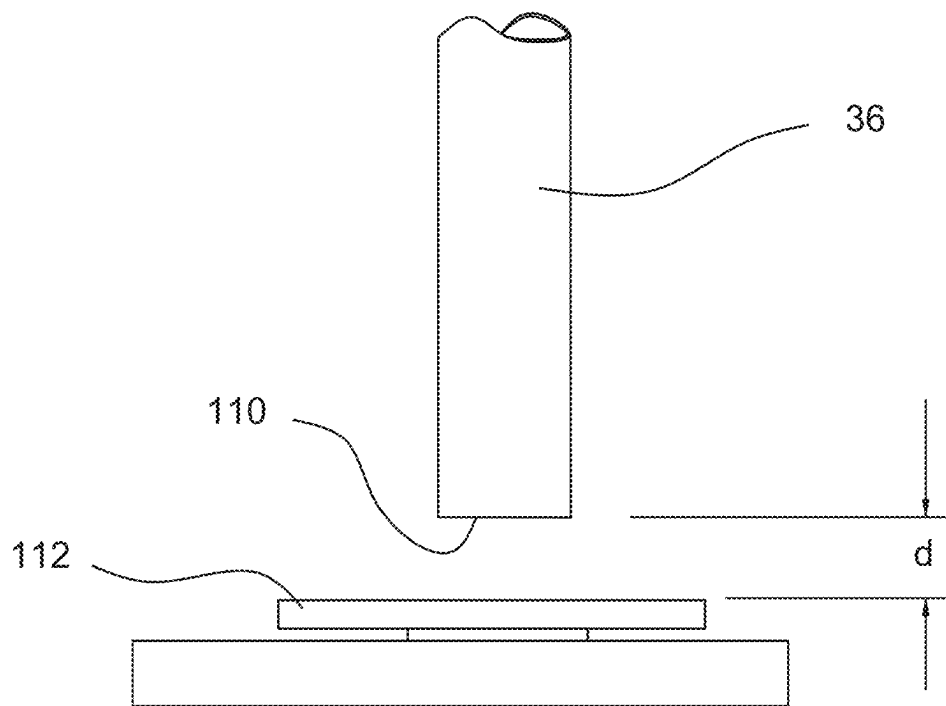


FIG. 4

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METHOD OF CONFIGURING AIR ACCELERATOR

CROSS-REFERENCE TO RELATED APPLICATION

This application is a divisional of U.S. patent application Ser. No. 17/340,184, filed Jun. 7, 2021, which is a divisional of U.S. patent application Ser. No. 15/973,860, filed May 8, 2018, which is a continuation application of U.S. patent application Ser. No. 13/546,649, filed Jul. 11, 2012, which claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Application No. 61/506,465, filed on Jul. 11, 2011, the entire contents of each which are incorporated herein by reference thereto.

FIELD OF THE DISCLOSURE

This disclosure generally pertains to apparatus for metering material that includes particles. More specifically, this disclosure concerns apparatus having a compressed air acceleration.

OVERVIEW

This disclosure has particular application to pouching machines used for forming and assembling pouches of particulate material, such as by way of example fine cut smokeless tobacco. Typical pouching machines simultaneously form and assemble, for example, ten pouches from a substantially continuous strip or web of pouch material and metered charges of prepared smokeless tobacco. To effect the simultaneous pouch assembly, pouching machines typically include a bank of generally vertical tobacco feed tubes. Typical pouching machines also include arrangements for drawing and directing a strip or ribbon of pouch web to each feed tube, and wrapping the strip around the corresponding feed tube to form a tubular formation, as well as arrangements to repetitively close and seal that tubular formation so as to form a lower transverse seam at a lower end portion of the tubular web formation just prior to charging each tubular formation with predetermined amount of smokeless tobacco. The pouching machine further includes arrangements for repetitively feeding individual charges of tobacco down corresponding feed tubes and into corresponding tubular formations. After each tobacco charge, the pouching machines close and seal the tubular formation at a second location above the tobacco charge to form an individual loaded and sealed pouch, which is then severed from the tubular formation.

Typically, smokeless tobacco material has a low moisture content, for example, about 30 to about 40% moisture level, and optionally includes flavorants, humectants and/or other tacky substances. Accordingly, smokeless tobacco has a tendency to stick to machine surfaces. Such smokeless tobacco is difficult to feed through pouch forming machines that rely merely on gravity feed techniques. Some pouching machinery incorporates pressurized air in the tobacco feed tubes to augment gravitational delivery of the smokeless tobacco charges. Because drier tobaccos are lighter than wetter tobaccos, the drier tobaccos have a greater tendency to scatter if subjected to jets of pressurized air during feeding, and that scatter can adversely affect the top seal on the associated pouch.

Prior pouching machines include a tobacco feed mechanism for repetitively discharging a predetermined amount of tobacco from a hopper or the like into a funnel at the upper

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end portion of a tobacco feed tube. Generally, if gravity is the only active force to move the tobacco down the feed tube, a charge of tobacco released into the tube forms into a column of tobacco traveling down the feed tube such that it is constrained along a significant path length that may be too long for proper filling operations. More particularly, not all of the entrained tobacco may have time to enter the confines of a partially closed pouch before the machine closes and seals the pouch along its upper transverse seam.

One solution has been to establish a Venturi arrangement at the base of the funnel. In this arrangement, pressurized air is introduced into the feed tube from a manifold through four to six or so small channels. Those small channels are fixed in size and may vary from tube to tube depending on machine tolerances and the like. Any clogging of one or more of the small channels tends to affect tobacco delivery for that feed tube in such a way that the bank of feed tubes performs inconsistently from one feed tube to another.

Another disadvantage of the foregoing arrangement that the small channels may impart a horizontal or transverse velocity component to the air being introduced through the small channels, with the result that some tobacco flow back may be caused.

It is desired to have the feed tubes of the bank of tobacco feed tubes operate consistently amongst one another so that filling operations across the entire bank are consistent with one another.

BRIEF DESCRIPTION OF THE DRAWINGS

The many innovative features and aspects of the present disclosure will be apparent to those skilled in the art when this specification is read in conjunction with the attached drawings wherein like reference numerals are applied to like elements and wherein:

FIG. 1 is a schematic view in partial cross section of tobacco dosing apparatus;

FIG. 1A is a partial cross-sectional view of the feed apparatus of FIG. 1;

FIG. 2 is an enlarged, partial cross-sectional view taken through the dose delivery apparatus of FIG. 1;

FIG. 3 is a detail view of the venturi discharge for the air accelerator unit of the dose delivery apparatus; and

FIG. 4 is a schematic illustration of a calibration set-up.

DETAILED DESCRIPTION

In the production of pouched products, including for example and without limitation, smokeless tobacco products, continuous-motion packaging machinery is often used, and is commonly known as form/fill/seal equipment. Such machinery receives packaging material is substantially continuous strips, receives material to be pouched as a substantially continuous supply from a supply chamber, meters substantially uniform quantities of the material, partially forms a pouch, fills the metered material into the pouch, and finally seals the pouch such that the pouch surrounds that material. While various companies make such equipment, one such company is known as Ropak.

Typical form/fill/seal equipment produces pouched products in a plurality of parallel streams of packaging material and product. For example, 5, 10, or more parallel lanes may be provided. Operating speeds on the order of 100 cycles per minute are known for each of the parallel lanes. As may be expected, that actual manufacturing speed depends on, for example, product flow characteristics, packaging materials used, and temperature at which filling occurs.

In accord with this disclosure, a form/fill/seal apparatus 10 typically includes a plurality of suitable dose delivery apparatuses 20 (see FIG. 1) to deliver particulate material in predetermined quantities. Typically, the form/fill/seal apparatus 10 receives a quantity of material to be parsed into predetermined quantities of doses of that material, and then delivers each predetermined quantity of material to a dose delivery apparatus 20. The dose delivery apparatus 20 moves the predetermined quantity of material to a portion of the form/fill/seal apparatus where a pair of continuous webs 22, 24 have been joined with a transverse seal 28 and longitudinal edge seals 26, 26' so as to define a pocket or pouch 29. That pocket or pouch 29 is typically formed around the end 30 of a discharge tube of the dose delivery tube of a corresponding dose delivery apparatus 20. Alternatively, a single web may be folded into a tubular form about the dose delivery tube and sealed along a single longitudinal edge, whereupon transverse seals applied to the tubular structure define a pouch 29. Since the dose delivery apparatuses 20 are essentially identical, it will suffice to describe one in detail, with it being understood that the others are substantially the same. The principal difference from one dose apparatus 20 to another resides in its connection with the supply conduit.

Each dosing apparatus 20 may include a supply conduit 24 connected at one end to the form/fill/seal apparatus 10 and connected at the other end to metering apparatus 12. The metering apparatus 12 is operable to receive particulate material from the apparatus 10, parse the particulate material into predetermined portions, doses, or quantities, and then deliver those predetermined portions, doses, or quantities of particulate material to the upper end of the dose delivery apparatus 20 at predetermined time intervals. The predetermined time intervals are selected so that a dose is delivered to the dose delivery apparatus 20 as each partial pouch is ready to be filled.

While the metering apparatus 12 may take a variety of physical forms and arrangements, a presently preferred arrangement is depicted in FIG. 1. More specifically, the metering apparatus 12 preferably includes a pair of generally parallel feed screws 14a, 14b that are arranged so as to be generally perpendicular to the axis of the dose delivery apparatus 20. A suitable conventional drive mechanism is connected to at least one of the feed screws 14a, 14b such that the two feed screws rotate in the same direction about their respective axes. The drive mechanism is controlled, in a conventional manner, such that the feed screws intermittently rotate, with the time interval of the intermittent rotation being operable to define the predetermined dose of particulate tobacco material delivered to the dose delivery apparatus 20.

The feed screws 14a, 14b are preferably designed such that the flight of one screw cleans the flight of the adjacent screw as the two screws rotate. This characteristic of the feed screws 14a, 14b helps assure consistent weight and volume for the predetermined doses being delivered to the dose delivery apparatus 20. Furthermore, the feed screws 14a, 14b are preferably fabricated from polyether ether ketone (PEEK).

The metering apparatus 12 also includes a housing 16 (see FIG. 1A) within which the feed screws 14a, 14b are positioned and within which those feed screws are mounted for rotation. The discharge end of the housing 16 is positioned above the inlet to the dose delivery apparatus 20, and may be offset from both the center and the edge as depicted so that particulate tobacco material of a given dose can drop directly in to dose delivery apparatus 20. The housing 16

closely conforms to the peripheral edge of the flight of each feed screw 14a, 14b so that particulate material does not spill over the edge of the flight and dosing quantity is thus controlled. Preferably, the housing 16 is also fabricated from PEEK.

The discharge end of the housing 16 is connected to a snout 18 which encloses the end of the housing and couples the housing 16 to the upper end of the funnel 32 of the dose delivery apparatus 20. The snout 18 assures that particulate tobacco material delivered to the dose delivery apparatus 20 by the feed screws 14a, 14b does not escape and falls into the dose delivery apparatus 20. In addition, the snout 18 is effective to avoid any external contamination of the particulate tobacco material passing therethrough. The snout 18 is also preferably fabricated from PEEK.

The use of PEEK as a preferred material for fabrication of the feed screws 14a, 14b, the housing 16, and the snout 18 has several advantageous and desirable attributes. PEEK functions as a thermal insulator. Thus, use of PEEK between the delivery apparatus 10 and the dose delivery apparatus 20 functions to substantially thermally insulated those apparatuses from one another. Moreover, PEEK substantially reduces and effectively avoids sticking of the particulate tobacco material to the surfaces of the housing, the feed screws, and the snout. Especially where the apparatus must be disassembled and cleaned on a regular basis (e.g., daily), this attribute is highly advantageous because it can reduce the cleaning time and thus add more processing time to the apparatus.

For purposes of this disclosure, the particulate material may be particulate tobacco that has optionally been blended with other components including, for example, flavorants, humectants, and/or other substances, some or all of which may be tacky or may add tackiness to the particulate tobacco. The particulate tobacco material may include fine cut tobacco that has been comminuted at about 70 cuts per inch. Preferred particulate tobacco material may include up to about 39% oven volatiles.

The snout 18 of the metering apparatus 12 attaches to a supply funnel 32 (see FIG. 1) at the inlet of the dose delivery assembly 20. Preferably, the supply funnel 32 is circularly symmetric about an axis passing therethrough. At the bottom end of the supply funnel 32, and in communication with the interior of the supply funnel, an air accelerator assembly 34 is provided. This air accelerator assembly 34 is operable to provide continuous or pulsed flow of particulate tobacco material. To that end, the air accelerator assembly 34 connects with an air supply conduit 38, which in turn receives pressurized air from an air supply 40. The air supply 40 may be a pump, air compressor, plenum chamber, or the like, as may be desired or appropriate for a particular application. A valve 42 may be in fluid communication with the air supply 40 and the air accelerator assembly 34. As desired, the valve 42 may be operable to interrupt air flow to the air accelerator assembly 34 so as to start, stop, and/or pulse air delivered to the air accelerator assembly 34. Typically, air at ambient temperature and pressure in the range of 4-18 psig has been found to be suitable for use with an air accelerator assembly 34 of the type described herein.

At the bottom end, the air accelerator assembly 34 attaches to a dosing tube 36. That dosing tube 36 preferably terminates in a position where the pouch has been partially formed and can receive particulate material from the discharge end of the dosing tube 36.

The air accelerator assembly 34 includes a body 50, and an internal member 52 which is axially adjustable with respect to the body 50 along an axis 54. Preferably, the

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funnel member 32 is rotationally symmetric about the axis 54. Internal surfaces of the body 50 that are exposed to air flow, as well as surfaces of the internal member 52 that are exposed to air flow or to product flow are also rotationally symmetric with respect to the axis 54.

The narrow or lower end of the funnel member 32 preferably includes a radially extending flange 56 having a periphery that corresponds to the outer peripheral surface of the body 50. In addition, the flange 56 of the funnel member 32 includes a radially extending annular face 64 which is configured to mate with a corresponding radially extending annular face 66 at the upper end of the body 50. The flange 56 preferably also includes a projecting land 68 which is received in a threaded bore 70 of the body 50. Cooperation between the projecting land 68 and the associated bore 70 assures that the body 50 and the funnel member 32 are coaxial when joined together. To that end, a plurality of axially extending bolts, or threaded fasteners 58, may be used to attach the flange 56 and the body 50. Suitable gasket material may be provided between the abutting surfaces 64, 66 of the flange 56 and the body 50, respectively, if desired.

Extending longitudinally through the body 50, along the axis 54, is a body cavity that includes a threaded, generally cylindrical portion adjacent the funnel member 32, a frustoconical portion 72 extending downstream from the threaded portion, and a discharge tube connection portion at the lower or bottom end of the body 50. The frustoconical portion 72 essentially matches the diameter of the threaded portion at its upstream end. In addition, the downstream or lower end of the frustoconical portion 72 is preferably sized to have a diameter corresponding to the inside diameter of the discharge tube 36. The discharge tube 36 is preferably attached to the downstream end of the body 50 using a suitable conventional attachment. For example, any of a threaded connection, a welded connection, or an adhesively bonded and sealed connection would be satisfactory.

Turning to the longitudinally movable or adjustable member 52 of the air accelerator assembly 34, the adjustable member 52 includes a generally cylindrical longitudinal bore 80 extending from the upstream end to the downstream end of the adjustable member 52. The longitudinal bore 80 preferably has a diameter corresponding to the opening at the discharge end of the funnel member 32 so that particulate material can move downwardly through the funnel member 32 and into the longitudinal bore 80 substantially free of impediment.

The upper or upstream end of the adjustable member 52 includes a flange portion 84 preferably having a peripherally threaded portion that mates with the threaded portion of the cavity in the body 50. Cooperation between the externally threaded flange 84 and the internally threaded portion of the body cavity not only secures the adjustable member 52 in the body 50, but also allows the adjustable member 52 to have its spatial relationship with the body 50 controlled in the longitudinal direction along the axis 54.

Preferably, the exterior surface of the adjustable member 52 also includes a frustoconical surface 82 extending from the flange 84 to the distal end 88 at the downstream end of the adjustable member 52. Preferably, the frustoconical surface 82 meets the longitudinal bore 80 at the distal end 88 of the adjustable member 52 so that an acute sharp angle is defined in the material of the adjustable member 52. Both the frustoconical surface 82 of the adjustable member 52 and the frustoconical portion of the cavity in the body 50 are preferably polished. Because the facing frustoconical surfaces define a chamber for pressurized air, and because it is desirable to accurately control the flow rate of pressurized

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air through that chamber, it is believed to be important that those facing frustoconical surfaces be as smooth as possible so as to avoid creating inconsistent resistance to air flow from one air accelerator assembly 34 to another. Accordingly, these facing frustoconical surfaces may be honed and/or polished so that the surface roughness is about 100 microinches or less, and preferably about 30 microinches or less.

As noted, the cavity of the body 50 and the frustoconical surface 82 of the adjustable member 52 cooperate to define a chamber 90 for pressurized air. That chamber 90 has fluid communication with the conduit 38, and thus the pump 40 and associated control valve 42 (see FIG. 1). The frustoconical surface 82 (see FIG. 3) of the adjustable member defines an angle α with the axis 54 of its central bore 80. The frustoconical surface portion 72 of the cavity in the body 50 has an angle β with the axis 54. The distal end 88 of the adjustable member 52 cooperates with the frustoconical surface portion 72 of the cavity in the body 50 to define a throat or minimum flow area at the downstream end of the chamber 90. To assure that the flow area through the chamber 90 decreases as air moves downstream toward the throat, the angle α must be less than the angle β . Thus, the chamber 90 (see FIG. 3) effectively comprises a venturi through which pressurized air in the chamber 90 passes as it moves toward and through the reduced area throat 100 (outlet of the chamber 90). With the longitudinal adjustability of the member 52 in the direction of the arrow 102, the throat 100 can be adjusted as described more fully below to calibrate and adjust the various air acceleration assemblies of a form/fill/seal machine.

Since it is also important that air supplied to the chamber 90 (see FIG. 2) through the conduit 38 be constrained to pass out of the chamber 90 only through the throat 100, a suitable conventional gasket 86 may be provided at the upper end of the chamber 90 between the flange 84 of the adjustable member 52 and the cavity of the body 50.

In a preferred embodiment, the body 50 and the adjustable member 52 are constructed from air-hardened tool steel.

As noted above, the particulate tobacco material processed through the dosing tube assembly described above may exhibit tackiness. Accordingly, one or more of the interior surface of the funnel member 32, the cylindrical channel 80 of the adjustable member 52, and the interior of the discharge tube 36 may also be coated with polyether ether ketone (PEEK). More preferably, the adjustable member 52 may be constructed entirely from PEEK. Such a coating can improve mechanical and chemical resistance to the particulate material as that material moves through the dosing tube assembly.

It will now be understood by those skilled in the art that the tapered angle β of the adjustable member 52 (see FIG. 2) is greater than the corresponding taper angle α of the frustoconical channel of the body 50 such that as the member 52 is threaded into the body 50 a tapered convergent chamber 80 is defined around a portion of the adjustable member 52 in the space provided between the body 50 and the member 52. As the member 52 is threaded further and further into the body 50, the annular discharge orifice or throat 100 at the distal end 88 of the member 52, and near the base of the body 50, becomes smaller and smaller.

Conventional set screws may be provided as a locking means to fix or otherwise lock the relative positions of the member 52 and the body 50.

To prepare an air acceleration assembly 34 for use, the assembly 34 and its discharge tube 36 are removed from the tobacco feed system. Then the assembly 34 is calibrated by

adjusting the throat of the variable venturi such that a predetermined force is obtained from the associated discharge tube. To that end, the assembly 34 with its discharge tube 36 is positioned in a fixture such that the end 36 at the base of the discharge tube 36 is proximately positioned relative to a suitable conventional a precision scale 112. The discharge tube 36 is held at a predetermined stand-off distance d above the surface of the precision scale 112. Preferably that predetermined stand-off distance d between the end of the discharge tube 36 and the precision scale 112 is about 1 mm.

The feed tube is connected to the source 40 of pressurized air through the conduit 38 (see FIG. 1) and the pressure regulator 42. The pressure regulator is adjusted to a desired operating pressure for the tobacco pouching machine, for example eighteen psig. Then the longitudinally adjustable member 52 is rotated so that it can be adjusted either up or down relative to the body 50 until the discharge of air through the discharge tube onto the precision scale registers a reading of a predetermined force, preferably in the range of about 20 to about 30 g. For example, the predetermined force or target scale reading might be 25 g. Once body 50 and member 52 have been adjusted so that the desired force reading is obtained, the member 52 is locked in place relative to the body 50 by a set screw or other suitable mechanism to fix the relative position of the body 50 and the member 52. While a mechanical locking arrangement such as a set screw may be used, the relative positions of the member 52 and the body 50 are most preferably permanently attached to one another, as by welding, so that the calibration is fixed. Otherwise, when the feed tube is cleaned (typically a daily occurrence), recalibration is required. The foregoing steps are repeated for each remaining air acceleration assembly 34 until all assemblies 34 have been calibrated to provide the same predetermined force.

After each air acceleration assembly 34 has been calibrated and returned to the tobacco feed mechanism, the pouching machine, i.e., the form/fill/seal machine, is ready for operation. Typically, a machine operator adjusts the air regulator 42 (FIG. 1) of the pouching machine to achieve desired pouch loading operation across the bank of feed tubes.

At one extreme, the air pressure may be too high, in which case the tobacco is driven into the pouch with such force that the pouch tends to open or cause tobacco to enter the first lower transverse seal of the pouch being formed. In another case, the pressure may be too low such that the upper transfer seam is closed and sealing initiated before all the tobacco has fully arrived into the body portion of the pouch. For this latter condition, the operator typically increases the operating pressure. Once the filling sequence has been optimized, the operator is assured uniform filling across the bank of feed tubes, because each air acceleration assembly has been calibrated the same way.

Preferably, the operating pressure of all feed lanes (or delivery apparatuses 20) is adjustable from a single, common regulator 42. Such arrangement contributes uniform tobacco feeding characteristics across the entire bank of feed lanes to enhance machine operation and performance. The arrangement assures that downstream timing requirements are uniformly met. For example the cutting knives for severing fully formed pouches operate uniformly at a fixed rate across the entire bank of feed lanes. The feed system as taught herein, with its locking down each air delivery system to a common, uniform calibration and uniform adjustment of operating pressure from a common regulator assures that tobacco is delivered at the right time and at the right speed

across the bank of feed lanes. During operations, should delivery speed of the feed lanes drift, the operator may return the entire bank of feed lanes back into desired delivery speed by observing a single feed lane while adjusting the common regulator.

In this description, the word “substantially” is used as an adjective to show that the modified term need not be used literally, but is intended to include equivalent terms which do not materially depart from the spirit and scope of the term. When the word “substantially” is used in connection with a geometric term, it is intended that the geometric term not be interpreted rigidly with respect to geometric definitions.

To similar effect, the word “about” is used in this description in connection with numerical terms to demonstrate that mathematical precision is not required and that a tolerance of $\pm 10\%$ around that numerical term is intended.

It will now be apparent to those skilled in the art that this specification provides a novel and unobvious improvement to a metering device for particulate material, particularly where pressurized fluid functions to assist movement of the particulate material through the apparatus. Furthermore, it will be apparent to those skilled in the art that numerous modifications, variations, substitutions, and legal equivalents exist for features of the invention described herein. Accordingly, it is expressly intended that all such modifications, variations, substitution, and legal equivalents that fall within the spirit and scope of the appended claims be embraced thereby.

What is claimed is:

1. A method, comprising:

first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity, the singular bore having a uniform cross-sectional flow area along a longitudinal length of the singular bore.

2. The method of claim 1, wherein the first configuring further includes

first establishing a test airflow through the chamber, measuring a first force of the test airflow exiting the first air accelerator, and

determining a first axial position of the adjustable member that causes the first force to equal a target force, the target force being a desired force that causes the first air accelerator to deliver the target quantity of the particulate material,

the chamber being defined in part by an external frustoconical surface of the adjustable member and internal frustoconical surface that is part of the interior surface, the internal frustoconical surface having a first angle of incline that is steeper than a second angle of incline of the external frustoconical surface relative to a first longitudinal centerline running through a center of the singular bore.

3. The method of claim 2, further comprising:

second establishing an operating airflow through the chamber with the adjustable member in the first axial position;

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third establishing a flow of the particulate material into the singular bore; and discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

4. The method of claim 1, further comprising:

first establishing an operating airflow through the chamber;

second establishing a flow of the particulate material into the singular bore; and

discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

5. The method of claim 4, wherein the second establishing further includes

connecting a funnel of a metering assembly to an entrance end of the body such that the funnel is in communication with the singular bore, a first internal diameter of an outlet of the funnel being equal to a second internal diameter of the singular bore, and

configuring the metering assembly to deliver the flow of the particulate material through the funnel and into the singular bore.

6. The method of claim 4, wherein the second establishing further includes

connecting a funnel of a metering assembly to an entrance end of the body such that the funnel is in communication with the singular bore, a first internal diameter of an outlet of the funnel being equal to a second internal diameter of the singular bore, and

configuring parallel feed screws of the metering assembly to transport the flow of the particulate material in a horizontal direction into and through the funnel and into the singular bore, the funnel having a first longitudinal centerline running through a center of the funnel that is positioned vertically and is aligned with a force of gravity, the first longitudinal centerline being aligned with a second longitudinal centerline running through a center of the singular bore.

7. A method, comprising:

first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity;

first establishing an operating airflow through the chamber;

second establishing a flow of the particulate material into the singular bore;

discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow;

inserting a dosing tube into a first end of a partially formed pouch, the singular bore being in communication with the dosing tube;

discharging the target quantity of the particulate material from the dosing tube into the partially formed pouch; and

sealing the first end.

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8. A method, comprising:

first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity;

first establishing an operating airflow through the chamber;

second establishing a flow of the particulate material into the singular bore; and

discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow,

wherein the second establishing further includes

connecting a funnel of a metering assembly to an entrance end of the body such that the funnel is in communication with the singular bore, a first internal diameter of an outlet of the funnel being equal to a second internal diameter of the singular bore, and configuring the metering assembly to deliver the flow of the particulate material through the funnel and into the singular bore.

9. The method of claim 8, further comprising:

inserting a dosing tube into a first end of a partially formed pouch, the singular bore being in communication with the dosing tube, a third internal diameter of the dosing tube being equal to the second internal diameter;

discharging the target quantity of the particulate material from the dosing tube into the partially formed pouch; and

sealing the first end.

10. A method, comprising:

first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity,

wherein the first configuring further includes

first establishing a test airflow through the chamber, measuring a first force of the test airflow exiting the first air accelerator, and

determining a first axial position of the adjustable member that causes the first force to equal a target force, the target force being associated with the target quantity of the particulate material, the chamber having a continuously smaller cross-sectional flow area from an air inlet to the air outlet of the chamber.

11. The method of claim 10, further comprising:

second establishing an operating airflow through the chamber with the adjustable member in the first axial position;

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third establishing a flow of the particulate material into the singular bore; and
 discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

12. The method of claim 11, further comprising:
 repeating the first establishing, the measuring and the determining steps on a periodic basis, following the discharging of the target quantity for a period of time.

13. The method of claim 11, further comprising:
 repeating the first configuring step for a plurality of additional air accelerators, each of the plurality of additional air accelerators being identical to the first air accelerator,

wherein the second establishing establishes the operating airflow by connecting a common air supply to each respective one of the chambers of each one of the plurality of additional air accelerators.

14. The method of claim 10, further comprising:
 inserting a gasket at an upper end of the chamber at an interface between the body and the adjustable member; and

connecting a flange to an upper end of the first air accelerator, a projecting land of the flange extending into the cavity, an annular gap existing between the projecting land and the adjustable member.

15. The method of claim 14, further comprising:
 second establishing an operating airflow through the chamber with the adjustable member in the first axial position;

third establishing a flow of the particulate material into the singular bore; and
 discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

16. The method of claim 14, further comprising:
 fixing the adjustable member in the first axial position prior to connecting the flange.

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17. A method, comprising:

first configuring a first cross-sectional flow area of a throat within a first air accelerator to discharge a target quantity of a particulate material, the first air accelerator including a body and an adjustable member, an interior surface of the body defining a cavity containing the adjustable member, the throat being at an air outlet of a chamber defined between the body and the adjustable member, the throat being near a discharge end of a singular bore defined by the adjustable member, the first configuring including axially moving the adjustable member along a first longitudinal length of the cavity,

wherein the first configuring further includes

first establishing a test airflow through the chamber, measuring a first force of the test airflow exiting the first air accelerator, and

screwing the adjustable member into a first axial position using first threads on an upper end of the adjustable member that mate with second threads on an upper end of the interior surface, the first axial position causing the first force to equal a target force, the target force being a desired force that causes the first air accelerator to deliver the target quantity of the particulate material,

the chamber having a continuously smaller cross-sectional flow area from an air inlet to the air outlet of the chamber.

18. The method of claim 17, further comprising:
 second establishing an operating airflow through the chamber with the adjustable member in the first axial position;

third establishing a flow of the particulate material into the singular bore; and discharging the target quantity of the particulate material from the singular bore with assistance from the operating airflow.

19. The method of claim 1, wherein the particulate material is smokeless tobacco.

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